WLC-Charge Receiver

WLC-Charge Rx

The WLC-Listener(WLC-L) has an integrated low power WLC-Charge Receiver(WLC-Charge Rx). WLC-L library includes a WLC-Charge Rx wrapper, based on **PCA943x** family of Charging ICs. **PCA943x** variants are :

- **PCA9431** Low power wireless charging receiver.
- **PCA9430** Low power wireless charging receiver with liner battery charger functions.

WLC-Charge Rx devices have an I2C interface for host micro controller or application processor to control the device features. WLC-Charge Rx device supports following features :

- High efficiency active rectifier with voltage regulations and protections.
- Temperature sensing by NTC pin(JEITA compliant charging).
- Auxiliary supply for dead battery.
- Linear battery charger/ LDO out.
- WLC-L(receiver) side interrupt function(RXIR) to WLC-P(transmitter) side.
- Antenna tuning.
- ADC to measure parameters for system control for system level safety.
- Comprehensive chip protections.

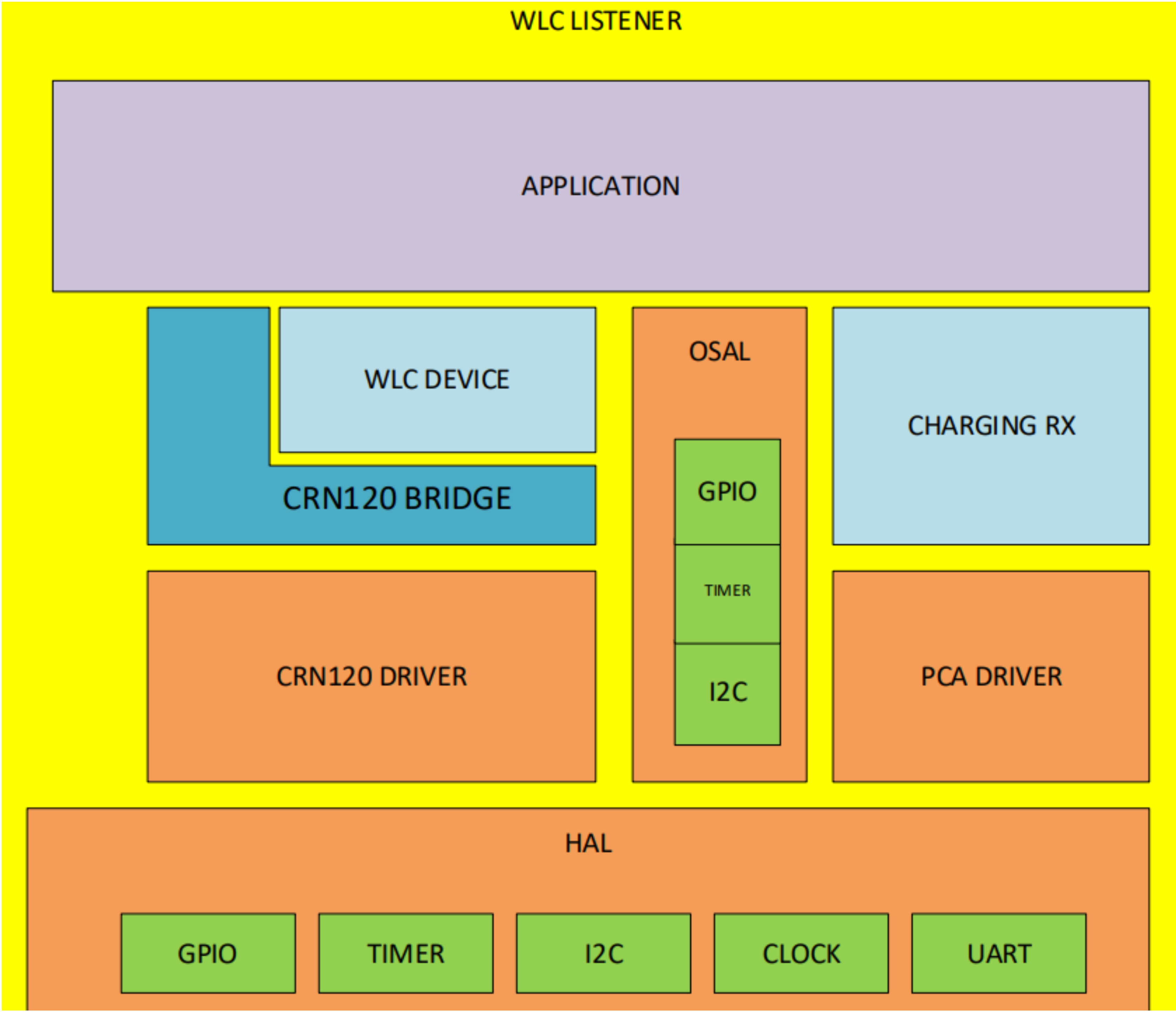
Refer WLC-Charge Receiver(**PCA943x**) data sheet for detailed functional description.

WLC-Charge Rx Wrapper

WLC-L library has an integrated WLC-Charge Receiver wrapper layer which provides APIs to check the status of WLC-Charge Rx and utilize its functions.This layer abstracts the underlying implementation of the WLC-Charge Rx driver(**PCA943x**) from the WLC-L application. To utilize the Charging Rx capabilities the WLC-L application needs to use the set of APIs provided by the WLC-Charge Rx wrapper.

WLC-L Library Software Stack

The following diagram illustrates the WLC-Charge Rx software stack organization in the WLC-L solution.


WLC LISTENER
APPLICATION
WLC DEVICE
CRN120 BRIDGE
OSAL
GPIO
TIMER
I2C
CHARGING RX
CRN120 DRIVER
PCA DRIVER
HAL
GPIO
TIMER
I2C
CLOCK
UART

WLC-L Firmware Structure

- **WLC-Charge Rx** : WLC-Charge Rx wrapper.
- **PCA943x** : Driver for PCA943x family of charging ICs.
- **WLC-Device** : NFC Forum TS compliant WLC-Device library.
- **CRN120** : Driver for CRN120 NFC Forum TS compliant tags.
- **HAL** : Hardware abstraction internal for WLC-Device library, mapped to OSAL APIs.
- **OSAL** : OS abstraction layer for WLC-L library. OSAL APIs need to be overridden by custom implementations for the host platform.
- **Peripheral Drivers** : Peripheral drivers of the host micro controller or application processor.
- **Charging Application** : Custom charging application using the APIs provided by WLC-L library stack.

WLC-Charge Rx Components

WLC-Charge Rx wrapper has the following components,

- **WLC-Charge Receiver** : Common WLC-Charge Rx functionalities. This component provides interfaces for initialize/de-initialize WLC-Charge Rx driver, enable/disable WLC-Charge Rx device, enable/disable and read interrupt status of WLC-Charge Rx device, and detect boot status of WLC-Charge Rx device.
- **WLC-Charge Receiver ADC** : This component provides interfaces for ADC integrated with WLC-Charge Rx device, to measure the parameters for system control needs. The APIs provided can enables direct measurement of internal voltages, currents, on-chip temperatures and off-chip temperatures.
- **WLC-Charge Receiver Battery Charger** : This component provides interfaces for linear battery charger functionalities available in wireless charging receiver. For linear battery charging functions, define **WLC_CHG_RX_BAT_CHG**. Linear battery charging functions are only available with **PCA9430** variants of WLC-Charge Rx ICs. To configure WLC-Charge Rx driver for linear battery charger, define **PCA9430_DRIVER**.
- **WLC-Charge Receiver VOUT** : This component provides interfaces for VOUT LDO functionalities for wirelerss charging receiver without linear battery charger functions. To configure WLC-Charge Rx driver without linear battery charger, define **PCA9431_DRIVER**. Don't define **WLC_CHG_RX_BAT_CHG**.
- **WLC-Charge Receiver RXIR** : This component provides interfaces for WLC-Charger Reciever side interrupt function to WLC-Charger Transmitter side(RXIR). When the reciever side AP or MCU want to notify WLC-Carger transmitter side, the AP or MCU can configure the RXIR registers and trigger the RXIR.
- **WLC-Charge Receiver Rectifier** : This component provides interfaces for input Rectifier of WLC-Charge Receiver. This rectifier converts the AC signal from the coil to a DC signal. These APIs supports configuration of under volatge lock-out and over voltage warning thresholds for WLC-

Charge Receiver input rectifier.

- **WLC-Charge Receiver VTUNE** : This component provides interfaces for WLC-Charge Receiver antenna tuning varactor. WLC-Charge Receiver generates a continuous VTUNE votlage to control external varactors, in order to have best resonance for power reciever coil.